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VW-PINNs: A volume weighting method for PDE residuals in physics-informed neural networks

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Physics-informed neural networks (PINNs) have shown remarkable prospects in solving the forward and inverse problems involving partial differential equations (PDEs). The method embeds PDEs into the neural network by calculating the PDE loss at a set of collocation points, providing advantages such as meshfree and more convenient adaptive sampling. However, when solving PDEs using nonuniform collocation points, PINNs still face challenge regarding inefficient convergence of PDE residuals or even failure. In this work, we first analyze the ill-conditioning of the PDE loss in PINNs under nonuniform collocation points. To address the issue, we define volume weighting residual and propose volume weighting physics-informed neural networks (VW-PINNs). Through weighting the PDE residuals by the volume that the collocation points occupy within the computational domain, we embed explicitly the distribution characteristics of collocation points in the loss evaluation. The fast and sufficient convergence of the PDE residuals for the problems involving nonuniform collocation points is guaranteed. Considering the meshfree characteristics of VW-PINNs, we also develop a volume approximation algorithm based on kernel density estimation to calculate the volume of the collocation points. We validate the universality of VW-PINNs by solving the forward problems involving flow over a circular cylinder and flow over the NACA0012 airfoil under different inflow conditions, where conventional PINNs fail. By solving the Burgers' equation, we verify that VW-PINNs can enhance the efficiency of existing the adaptive sampling method in solving the forward problem by three times, and can reduce the relative L_2 error of conventional PINNs in solving the inverse problem by more than one order of magnitude.

Physics-informed neural networks, Partial differential equations, Nonuniform sampling, Residual balancing, Deep learning

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1. Introduction

In recent years, physics-informed neural networks (PINNs) [1] have become a research hotspot for solving the forward and inverse problems involving partial differential equations (PDEs). The central idea of PINNs is integrating the governing equations of physical systems into the loss function of neural networks, ensuring that the neural networks minimize PDE residuals while approaching the definite conditions or observed data. In fact, the concept of PINNs can be traced back to the 20th century when Dissanayake and Phan-Thien [2] pioneered the use of neural networks to solve PDEs. Compared with mesh-based numerical methods such as finite element (FEM), finite difference (FDM), and finite volume (FVM), the advantage of PINNs lies in their ability to directly calculate spatiotemporal derivatives through automatic differentiation [3], enabling a meshfree approach that avoids mesh generation, which also indicates that PINNs can alleviate the curse of dimensionality to a certain extent [4,5]. Moreover, PINNs can conveniently integrate various observed data to solve inverse problems, such as inferring velocity and pressure fields based on concentration field and the Navier-Stokes equations [6]. In contrast, nu-

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merical methods require extremely high cost to achieve the same process. PINNs have been demonstrated for various physics phenomena, including fluid mechanics [6-9], heat transfer [10,11], fluid-structure interaction [12,13], electromagnetic propagation [14], and quantum chemistry [15].

Although PINNs have achieved gratifying performance, they still face challenges in terms of accuracy, computational efficiency and training robustness, especially for complex problems [16]. Therefore, over the past few years, numerous researchers have enhanced PINNs in various aspects. For example, adaptive activation functions have been designed to improve the convergence of the algorithm [17], and *hp*-VPINNs based on the weak form of PDE have been proposed to perform integral transformation on the PDE residuals through Legendre basis function [18]. gPINNs have been developed to embed gradient information of PDE residuals into the loss function [19]. For complex spatiotemporal problems, parallel frameworks [20,21] have been built based on spatiotemporal domain decomposition to accelerate the training of PINNs and improve the accuracy. To enhance multi-scale recognition capabilities, Fourier feature network [22] that scales spatiotemporal coordinates in sinusoidal space and MscaleDNN [23] that constructs multi-scale inputs a priori were proposed. Furthermore, the gradient pathologies in PINNs have been addressed through adaptive weighting [24-26] and hard constraints [27-29].

In PINNs, PDE loss is evaluated at a set of scattered collocation points. The effect of collocation points on PINNs is similar to the effect of mesh points on FEM. Thus, the location and distribution of these collocation points are highly important to the performance of PINNs [30]. When the solution of the PDE is a simple continuous smooth function, uniform sampling of the collocation points is usually appropriate. However, for many practical problems, such as flow over an object and shock wave capture in fluid mechanics, the gradients of the solution near the object and shock wave are large, while they are small in other computational regions. Thus, refining the collocation points near the object and shock wave to better capture the details of flow while maintaining low-density sampling in other regions can consider both computational cost and accuracy, which is commonly acknowledged in solving such problems by numerical methods. At present, although some studies on PINNs have paid attention to the importance of nonuniform sampling [30-39], these studies still calculate the mean square error of PDE residuals at all collocation points as the PDE loss. This loss evaluation method overlooks the inconsistent convergence of PDE residuals at different locations within the computational domain caused by differences in sampling density. Specifically, in the PDE loss of PINNs, the proportion of PDE residuals at all collocation points is equal. The reduction of local PDE residuals within high-density regions brings greater benefits of loss reduction compared with those within low-density regions when solving problems using nonuniform collocation points. The network naturally focuses on reducing the PDE residuals in these high-density sampling regions. Conversely, the residuals in low-density sampling regions have difficulty converging. Consequently, it diminishes the efficiency of PINNs and may even lead to failure, especially when there is a significant difference in sampling density. We refer to this issue as the ill-conditioning of the PDE loss function.

To address the issue, this work defines the volume weighting residual and proposes volume weighting physics-informed neural networks (VW-PINNs). The PDE loss function is re-evaluated by weighting the PDE residuals based on the volume occupied by the collocation points within the computational domain. To calculate the volume of collocation points in meshfree scenarios such as VW-PINNs, we also develop an efficient volume approximation algorithm based on kernel density estimation [40,41]. By solving four forward problems and one inverse problem, we validate the advantages of the proposed method in terms of universality, convergence, and accuracy.

The remainder of the paper is organized as follows. In Sect. 2, we provide a detailed introduction to PINNs and analyze the ill-conditioning of the PDE loss function in PINNs. Then, we propose VW-PINNs. In Sect. 3, we carry out a series of numerical experiments to validate the effectiveness of VW-PINNs. Concluding remarks and direction for future research are then presented in Sect. 4.

2. Methodology

2.1 PINNs

Differing from numerical methods, PINNs transform solving PDEs into an optimization problem. Their inputs usually involve spatiotemporal coordinates $(\mathbf{x},t)$, corresponding to the spatiotemporal variables of the PDEs, with the output being the solution $u(\mathbf{x},t)$. We consider the PDE defined on a domain $\Omega \subset \mathbb{R}^d$:

$$\begin{cases} u_t + \mathcal{N}[u] = 0, & \mathbf{x} = (x_1, x_2, ..., x_d) \in \Omega, t \in [0,T], \\ u(\mathbf{x},0) = g(\mathbf{x}), & \mathbf{x} \in \Omega, \\ \mathcal{B}[u] = 0, & \mathbf{x} \in \partial\Omega, t \in [0,T], \end{cases} \tag{1}$$

where u denotes the solution of the PDE, $\mathcal{N}[\cdot]$ is the general differential operator which can include any combination of linear and nonlinear terms of spatial derivatives, and $\mathcal{B}[\cdot]$ is the boundary condition operator. The initial condition at $t = 0$ is defined by $g(x)$.

The loss function of PINNs for this problem is defined as follows:

$$\mathcal{L} = \omega_{ic}\mathcal{L}_{ic} + \omega_{bc}\mathcal{L}_{bc} + \omega_r\mathcal{L}_r + \omega_{da}\mathcal{L}_{da}, \tag{2}$$

where

$$\mathcal{L}_{ic} = \frac{1}{N_{ic}} \sum_{i=1}^{N_{ic}} \left| u_\theta(\mathbf{x}_{ic}^i, 0) - g(\mathbf{x}_{ic}^i) \right|^2, \tag{3}$$

$$\mathcal{L}_{bc} = \frac{1}{N_{bc}} \sum_{i=1}^{N_{bc}} \left| \mathcal{B}\left[u_\theta(\mathbf{x}_{bc}^i, t_{bc}^i) \right] \right|^2, \tag{4}$$

$$\mathcal{L}_{r} = \frac{1}{N_{r}} \sum_{i=1}^{N_{r}} \left| \frac{\partial u_\theta}{\partial t}(\mathbf{x}_{r}^i, t_{r}^i) + \mathcal{N}\left[u_\theta(\mathbf{x}_{r}^i, t_{r}^i) \right] \right|^2, \tag{5}$$

$$\mathcal{L}_{da} = \frac{1}{N_{da}} \sum_{i=1}^{N_{da}} \left| u_\theta(\mathbf{x}_{da}^i, t_{da}^i) - u(\mathbf{x}_{da}^i, t_{da}^i) \right|^2. \tag{6}$$

In Eqs. (3)-(6), θ represents the network parameters, while $u_\theta(\mathbf{x}, t)$ denotes the solution of PINNs. $\{\mathbf{x}_{ic}^i, 0\}_{i=1}^{N_{ic}}$, $\{\mathbf{x}_{bc}^i, t_{bc}^i\}_{i=1}^{N_{bc}}$ and $\{\mathbf{x}_{r}^i, t_{r}^i\}_{i=1}^{N_{r}}$ are training points for the initial conditions, boundary conditions and PDE residuals, typically sampled directly within the computational domain. $u(\mathbf{x}_{da}^i, t_{da}^i)$ represents available observed data obtained through experiment or other methods, and the corresponding loss function Eq. (6) only exists in solving the inverse problems. During the network training, $\mathcal{L}_{ic}$, $\mathcal{L}_{bc}$ and $\mathcal{L}_{da}$ ensure the network outputs approach the initial conditions, boundary conditions and observed data, while $\mathcal{L}_r$ constrains them to satisfy the governing equation. The conventional PINNs framework is illustrated in Fig. 1.

The weight parameters ω_{ic}, ω_{bc}, ω_r and ω_{da} control the balance between different components in the loss function. The appropriate weight can speed up the convergence of PINNs training [42]. Due to the requirement for computing high-order derivatives in PINNs, the activation functions should be highly differentiable, with commonly used options including hyperbolic tangent function (tanh) and sine function (sin). Optimization algorithms usually employ Adam [43] and L-BFGS [44].

2.2 Ill-conditioning of the PDE loss function in PINNs caused by nonuniform sampling

At present, most researches on PINNs generally compute the PDE loss according to Eq. (5), which means that the proportion of PDE residuals at all collocation points is equal in the loss. It is reasonable when employing uniform sampling for collocation points because the distribution pattern of the collocation points is consistent with their proportion in the loss. However, when employing nonuniform sampling, due to disparities in sampling density among collocation points at different locations in the computational domain, computing the loss according to Eq. (5) will result in the dominance of PDE residuals from regions with high sampling density during network training. The network parameters

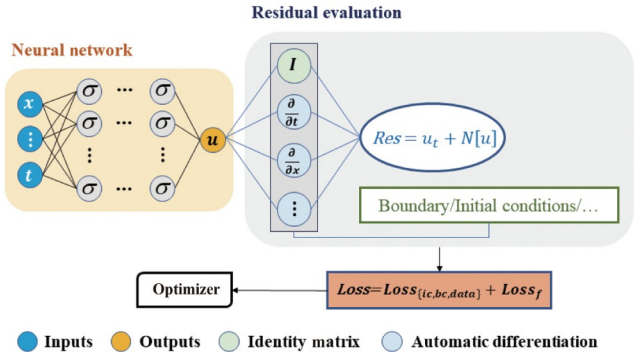


Figure 1 A schematic of PINNs for solving PDEs.

tend to sufficiently reduce the residuals in these regions, while other low-density regions receive little attention and the residuals converge difficultly. In other words, the reduction of conventional PDE loss cannot guarantee the convergence of the total PDE residual across the entire computational domain. The reason is as follows: considering the high-density and low-density sampling regions that occupy the same volume within the computational domain, it is evident that the former contains a greater number of collocation points than the latter. Thus, when the PDE residuals at all collocation points have an equal proportion in the PDE loss, the loss reduction benefits brought by reducing the local residuals in high-density regions are higher than those in low-density regions. Inevitably, the networks naturally focus on local learning in these high-density regions. However, as is well known, solving PDEs requires an adequate reduction in the residuals throughout the entire computational domain. According to the error propagation theory, erroneous results in regions with high residuals can lead to inaccuracies in the solution of other regions, even if the residuals in those regions are low [38]. Therefore, for nonuniform sampling problems, conventional PDE loss exhibits limitations, diminishing the solving efficiency of PINNs and potentially resulting in solving failure, especially for complex nonlinear problems with significant differences in sampling density.

We take the flow over an object in fluid mechanics as an example to illustrate the above issue, which is a typical nonuniform sampling problem. Because the gradient of the flow field is large near the object, a sufficiently dense set of collocation points is required to capture the flow details better. In contrast, the gradient in other regions is smaller, and sparse collocation points are adequate, which is commonly acknowledged in solving such problems. However, current research employing PINNs to solve the forward problems involving flow over an object commonly relies on nearly uniform collocation points within the channel [45,46] or within a specific local region [47-49]. In these cases, PDEs are solved within small computational domains, which are only several times the size of the objects. Thus, it

is acceptable to use dense collocation points in regions far from the object. However, for classical external flow problems at subsonic or transonic speeds, the far-field boundary is infinite in principle and is at least dozens of times larger than the object in practice. In such scenarios, uniformly sampling according to the resolution requirement near the object would result in an enormous number of collocation points, leading to significantly high computational costs in solving problems. Therefore, nonuniform sampling is necessary. We use PINNs to solve the inviscid compressible flow over a circular cylinder with the collocation points obtained by nonuniform sampling. The dimensionless governing equations for this problem are

$$\begin{aligned}
&\rho\left(\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}\right)+u\frac{\partial \rho}{\partial x}+v\frac{\partial \rho}{\partial y}=0,\\
&\rho\left(u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}\right)+\frac{\partial p}{\partial x}=0,\\
&\rho\left(u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}\right)+\frac{\partial p}{\partial y}=0,\\
&\rho c_v\left(u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}\right)+\gamma^2 p\left(\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}\right)=0.
\end{aligned} \tag{7}$$

In Eq. (7), u denotes the x-component of the velocity field, v the y-component. ρ, T and p represent density, temperature and pressure, respectively, satisfying the relationship: $p=\rho T/(\gamma Ma^2)$ (Ma is the Mach number). c_v is the specific heat at constant volume, $\gamma=1.4$ is the specific heat ratio, satisfying the relationship: $c_v=\gamma/[(\gamma-1)Ma^2]$. We set $Ma=0.4$, with $[x,y]$ as the network inputs and $[\rho,u,v,T]$ as the network outputs.

The cylinder is placed at $(x,y)=(0,0)$ with diameter $D=1$, while the far field is positioned at the same center with a diameter of 40. The computational domain and the distribution of collocation points are shown in Fig. 2. The total number of collocation points is $N_r=4800$. Meanwhile, we randomly select $N_{bc}=80$ points on the cylinder and enforce the non-penetrating condition $\mathbf{V}\cdot\mathbf{n}=0$. In the far field, we similarly select $N_{bc}=80$ points randomly and specify the dimensionless variables as constants $[\rho_\infty,u_\infty,v_\infty,T_\infty]=[1,1,0,1]$ (because the flow away from the object approaches the uniform freestream). We solve this problem using conventional PINNs with 5 hidden layers and 64 neurons per layer, tanh is chosen as the activation function. For the network training, we first run 3000 steps using the Adam optimizer, followed by an additional 2000 steps using the L-BFGS optimizer (maximum inner iterations is 20). The wall time of PINNs is 2.68 min.

Figure 3 displays the convergence of loss function obtained by PINNs. Evidently, both the boundary loss and the PDE loss are reduced by more than 5 orders of magnitude, which usually means that the networks obtain acceptable results. However, the flow field given by PINNs is wrong, as shown in Fig. 4. The distribution of PDE residuals shown in Fig. 5 explains the reason for the failure in solving. While the residuals near the cylinder with high-density sampling converge to the order of 10^{-4}, the residuals in the far-field freestream region with low-density sampling only drop to the order of 10^{-1} (although the residuals in the far-field freestream region are large, the averaged PDE loss remains low due to the small number of collocation points in this region), which indicates that the total PDE residual across the entire computational domain insufficiently converges. The information from the far-field freestream fails to propagate to the cylinder, resulting in an inevitable failure of the solving. Our perspective receives strong validation from this result.

2.3 VW-PINNs

To address the issue described in Sect. 2.2, based on the volume occupied by collocation points within the computational domain, we define the volume weighting residual. Taking Eq. (1) as an example, the volume weighting residual

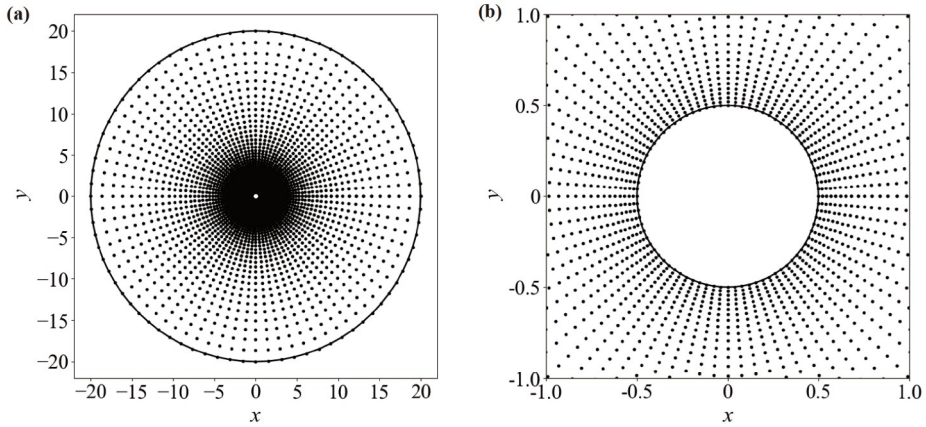


Figure 2 Computational domain and the distribution of collocation points in solving inviscid compressible flow over a circular cylinder. (a) Entire computational domain. (b) Near the cylinder region.

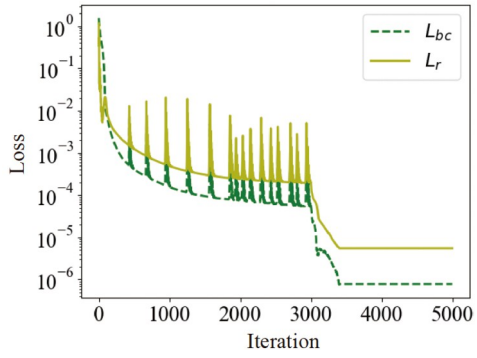


Figure 3 Convergence of loss function obtained by PINNs for solving inviscid compressible flow over a circular cylinder.

at collocation point (x_r^i, t_r^i) is

$$Res_v = \left\{ \frac{\partial u_\theta}{\partial t}(\mathbf{x}_r^i, t_r^i) + \mathcal{N}\left[u_\theta(\mathbf{x}_r^i, t_r^i)\right] \right\} V(\mathbf{x}_r^i, t_r^i), \tag{8}$$

where $V(\mathbf{x}_r^i, t_r^i)$ is the volume occupied by $(\mathbf{x}_r^i, t_r^i)$ in the computational domain. Compared with only calculating the residual of the PDE at $(\mathbf{x}_r^i, t_r^i)$, we weight the residual of PDE through the volume occupied by collocation points in the computational domain, embedding explicitly the distribution characteristics of collocation points in the residual evaluation. Building on this, we propose VW-PINNs, establishing a novel method for evaluating the PDE loss function. Still using Eq. (1) as an example, the PDE loss of VW-PINNs is

$$\mathcal{L}_r = \frac{\sum_{i=1}^{N_r} \left| \left\{ \frac{\partial u_\theta}{\partial t}(\mathbf{x}_r^i, t_r^i) + \mathcal{N}\left[u_\theta(\mathbf{x}_r^i, t_r^i)\right] \right\} V(x_r^i, t_r^i) \right|^2}{\sum_{i=1}^{N_r} \left[V(\mathbf{x}_r^i, t_r^i) \right]^2}. \tag{9}$$

Compared with the PDE loss of conventional PINNs, VW-PINNs impose PDE constraint through volume weighting residual. The residual of PDE at each collocation point is weighted by the volume it occupies, and the weight satisfies a linear inverse relationship with the sampling density of collocation point. Therefore, our method balances the differences in residual convergence across regions with various sampling densities in network optimization. In other words, the reduction of volume weighting PDE loss can ensure effective convergence of the total PDE residual across the entire computational domain. The convergence level of the volume weighting PDE loss determines the convergence status of the total PDE residual. In addition, it is obvious that when uniform sampling is employed, $V(\mathbf{x}_r^i, t_r^i) = V_s/N_r$ (V_s is the volume of the entire computational domain), and Eq. (9) degenerates into Eq. (5).

Since VW-PINNs are meshfree, it cannot compute volume relying on mesh topology as mesh-based numerical methods. Therefore, based on kernel density estimation method [40,41], we develop a volume approximation algorithm suitable for random sampling, allowing for the approximate calculation of volume at extremely low time cost. Kernel density estimation is a non-parametric statistical method. Its key concept involves placing a kernel function centered at each sampling point, such as the Gaussian kernel function utilized in this study. Subsequently, all these kernel functions are summed to estimate density. This method is widely applied in fields such as data analysis and machine learning.

Before stating the volume approximation algorithm, it is important to note that PINNs solve the PDE in the entire spatiotemporal dimensions directly. Thus, we choose to incorporate the temporal coordinate into the volume calculation. Using collocation points $\left\{\mathbf{x}_r^i, t_r^i\right\}_{i=1}^{N_r}$ as the center of Gaussian kernel functions, the probability density function $q(\mathbf{x}, t)$ in the computational domain can be obtained by

$$q(\mathbf{x}, t) = \frac{1}{N_r} \sum_{i=1}^{N_r} \exp\left(-\frac{\left\|\mathbf{x} - \mathbf{x}_r^i\right\|^2 + \left\|t - t_r^i\right\|^2}{2h^2} \right), \tag{10}$$

where $h > 0$ is the bandwidth, which controls the radial scope of the kernel functions. The probability density $q(\mathbf{x}_r^i, t_r^i)$ at each collocation point can be obtained by Eq.

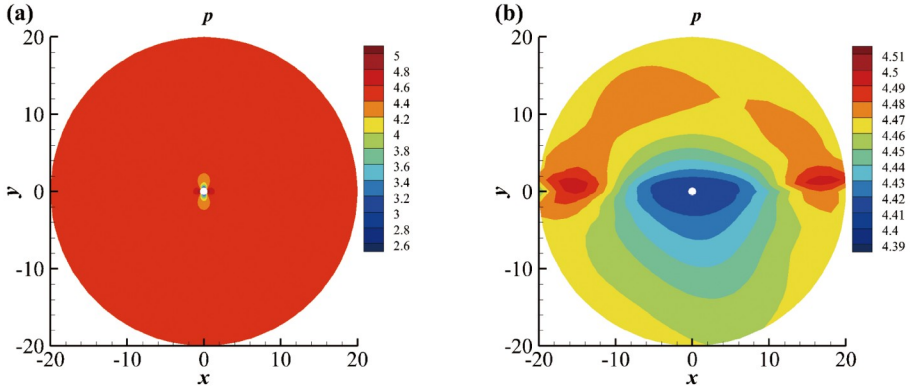


Figure 4 Comparison of pressure field between (a) the reference solution from FVM and (b) the solution given by PINNs for solving inviscid compressible flow over a circular cylinder.

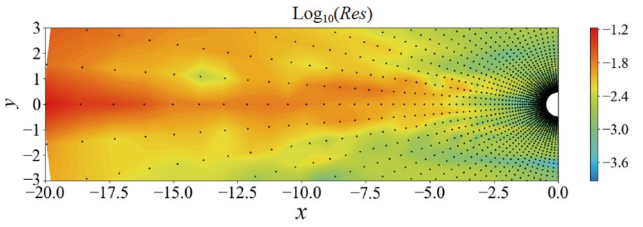


Figure 5 Residual distribution of Euler equations (7) given by PINNs for solving inviscid compressible flow over a circular cylinder.

(10). Calculate its reciprocal by

$$a(\mathbf{x}_r^i, t_r^i) = \frac{1}{q(\mathbf{x}_r^i, t_r^i)}, \tag{11}$$

and normalize $a(\mathbf{x}_r^i, t_r^i)$ by

$$a_{\text{norm}}(\mathbf{x}_r^i, t_r^i) = \frac{a(\mathbf{x}_r^i, t_r^i)}{\sum_{i=1}^{N_r} a(\mathbf{x}_r^i, t_r^i)}. \tag{12}$$

Finally, by multiplying a_{norm} with the total volume V_s of the entire computational domain, the approximate value of the volume occupied by each collocation point in the computational domain is obtained. A schematic illustration of VW-PINNs is shown in Fig. 6.

$$V(\mathbf{x}_r^i, t_r^i) = V_s a_{\text{norm}}(\mathbf{x}_r^i, t_r^i). \tag{13}$$

3. Results

We validate the effectiveness of VW-PINNs by solving four forward problems and one inverse problem. The forward problems include: inviscid compressible flow over a circular cylinder, viscous incompressible flow over a circular cylinder, viscous incompressible flow over the NACA0012 airfoil at the angle of attack $\alpha = 5°$, and Burgers' equation based on adaptive sampling [30]. The inverse problem involves Burgers' equation. To evaluate the accuracy of the solution, we calculate the relative L_2 error:

$$\frac{\|U_\theta - U\|_2}{\|U\|_2}, \tag{14}$$

where U_θ is the predicted solution by the neural network, U represents the numerical solution. For three flow cases, U is obtained through FVM, whereas for the Burgers' equation, we use the spectral method to obtain numerical solution. The study develops programs based on the PyTorch platform and executes the algorithm on NVIDIA GeForce RTX 3080 GPU.

3.1 Inviscid compressible flow over a circular cylinder

We first validate the effect of VW-PINNs by solving the inviscid compressible flow over a circular cylinder as described in Sect. 2.2. The governing equations are Eq. (7). VW-PINNs maintain the same solving settings as PINNs.

Figure 7 shows the comparison of loss function obtained by PINNs (Sect. 2.2) and VW-PINNs respectively. From the curves of the training loss, we observe that the boundary loss of both is reduced to the order of 10^{-6}, indicating that the approximation error of the boundary conditions is adequately low. In Fig. 7(b), the blue solid line represents the conventional PDE loss (Eq. (5)) used to gradient backpropagation in PINNs. The black dashed line represents the evaluation of volume weighting PDE loss (Eq. (9)) during PINNs training, which is only used for evaluation and not for gradient backpropagation (As we describe in Sect. 2, the volume weighting PDE loss is a more reasonable indicator to reflect the convergence level of PDE residuals in the entire computational domain for the problems of using nonuniform collocation points, because it avoids the influ-

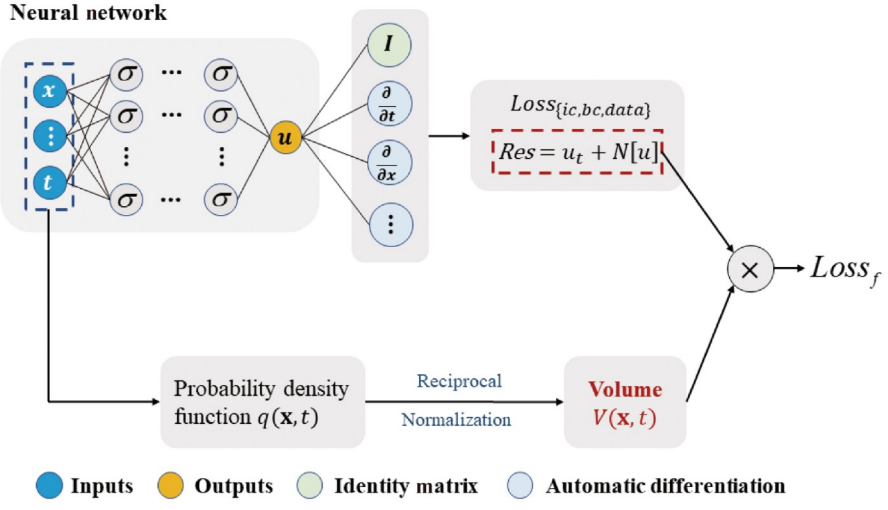


Figure 6 A Schematic of VW-PINNs for solving PDEs.

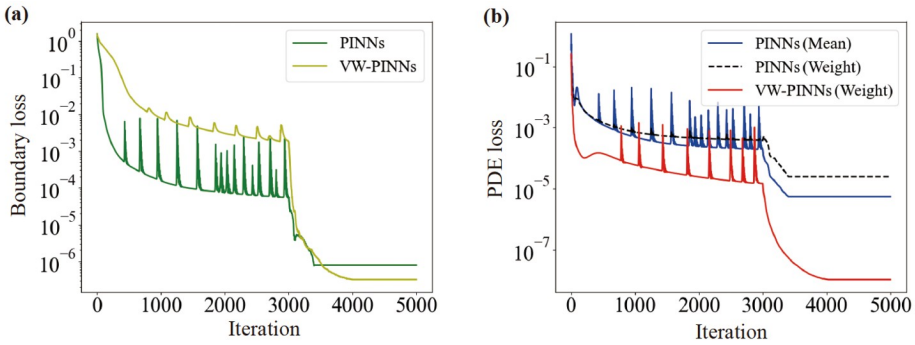


Figure 7 Comparison of loss function obtained by PINNs (Sect. 2.2) and VW-PINNs for solving inviscid compressible flow over a circular cylinder, respectively. (a) Boundary loss. (b) PDE loss. The blue solid line represents the conventional PDE loss (Eq. (5)) used to gradient backpropagation in PINNs. The black dashed line represents the evaluation of volume weighting PDE loss (Eq. (9)) during PINNs training, which is only used for evaluation and not for gradient backpropagation. The red solid line represents the volume weighting PDE loss used to gradient backpropagation in VW-PINNs.

ence of the differences in sampling density of collocation points). Obviously, the volume weighting PDE loss of PINNs is only reduced by 3 orders of magnitude. Corresponding to the residual distribution (Fig. 5), the total residual of the Euler equations does not sufficiently converge. In contrast, the volume weighting PDE loss of VW-PINNs is reduced by 7 orders of magnitude, resulting in an efficient convergence of the total PDE residual. In Fig. 8, we plot the pressure field near the cylinder given by VW-PINNs, which is consistent with the reference solution. For the problem of flow over an object, the pressure coefficient $C_p = (p - p_\infty)/[0.5\rho_\infty(u_\infty^2 + v_\infty^2)]$ on the object is usually of greater interest to researchers. Therefore, we use it to assess the accuracy of the solution, as shown in Fig. 9. The relative L_2 error between the pressure coefficient obtained by VW-PINNs and the FVM is only 1.33%, while the result obtained by PINNs is obviously incorrect. The wall time of VW-PINNs is 6.07 min. It is greater than the wall time of PINNs (Sect. 2.2). The reason is that PINNs quickly converge to wrong results and then encounter stagnation.

In Fig. 10, we plot the distribution of equation residuals given by VW-PINNs. Compared with the result given by PINNs (Fig. 5), our method achieves consistent and sufficient convergence of the residuals in the entire computational domain.

3.2 Viscous incompressible flow over a circular cylinder

Here we test VW-PINNs by solving the viscous incompressible flow over the circular cylinder at the Reynolds number $Re = 40$. The dimensionless governing equations of this problem are

$$\begin{aligned} &\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \\ &u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + \frac{\partial p}{\partial x} - \frac{1}{Re}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = 0, \\ &u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + \frac{\partial p}{\partial y} - \frac{1}{Re}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) = 0, \end{aligned} \tag{15}$$

where u denotes the x-component of the velocity field, and v denotes the y-component. p represents pressure. For $Re = 40$, there exist two symmetric separated vortices behind the cylinder. We take $[x, y]$ as the network inputs and

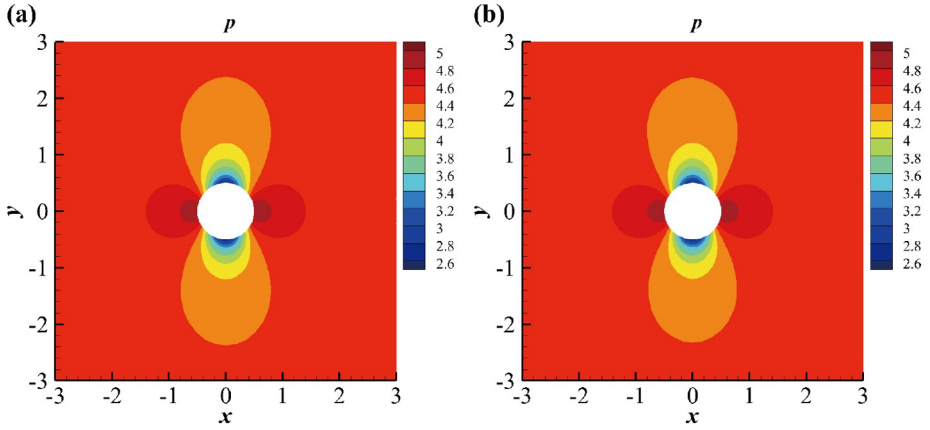


Figure 8 Comparison of pressure field near the cylinder between (a) the reference solution from FVM and (b) the solution given by VW-PINNs for solving inviscid compressible flow over a circular cylinder.

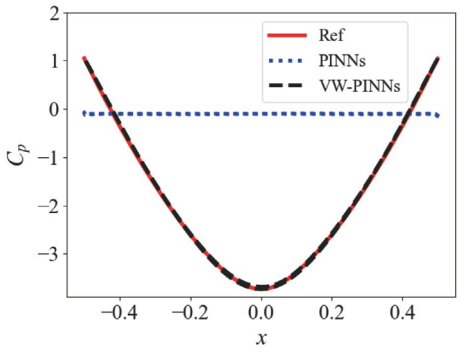


Figure 9 Comparison of pressure coefficient on the cylinder obtained by FVM (red), PINNs (blue) and VW-PINNs (black) for solving inviscid compressible flow over a circular cylinder, respectively.

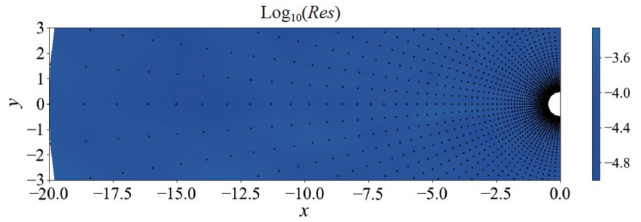


Figure 10 Residual distribution of Euler equations (7) given by VW-PINNs for solving inviscid compressible flow over a circular cylinder.

$[u, v, p]$ as the network outputs.

The cylinder is placed at $(x, y) = (0, 0)$ with diameter $D = 1$. The computational domain and the distribution of collocation points are illustrated in Fig. 11. The total number of collocation points is $N_r = 12300$. At the cylinder, inflow and outflow boundaries, we randomly place $N_{bc} = 200$, 100 and 75 points respectively, and impose no-slip condition $\mathbf{V} = \mathbf{0}$, velocity inlet condition $[u_\infty, v_\infty] = [1, 0]$ and pressure outlet condition $p_\infty = 0$. For the upper and lower boundaries, we randomly select $N_{bc} = 100$ points and also enforce velocity inlet condition. We solve the problem through PINNs and VW-PINNs respectively. The neural network contains 5 hidden layers with 64 neurons per layer, and tanh is chosen as the activation function. For the network training, we first run 3000 steps using the Adam optimizer, followed by an additional 2000 steps using the L-BFGS optimizer (maximum inner iterations is 20).

Figure 12 shows the comparison of the flow field near the cylinder among the reference solution, and the solutions given by PINNs and VW-PINNs. Obviously, PINNs fail to solve, while the results given by VW-PINNs are reasonable. The convergence of loss function obtained by two methods indicates the reason for their different results, as shown in Fig. 13. When the boundary losses of both are low enough, the volume weighting PDE loss of PINNs is reduced by only two orders of magnitude, meaning that the total PDE residual in the entire computational domain does not sufficiently converge. By comparison, VW-PINNs reduce the volume weighting PDE loss by 5 additional orders of magnitude. In Fig. 14, we evaluate the accuracy of the solution by the pressure coefficient on the cylinder. Using the solution obtained by the FVM as a reference, the relative L_2 error of VW-PINNs is only 0.53%, whereas PINNs yield incorrect result. The wall times of PINNs and VW-PINNs are 7.33 min and 17.94 min, respectively. The shorter wall time of PINNs is because they quickly converge to wrong results and then encounter stagnation.

In addition, we compare the streamline distribution near the cylinder given respectively through FVM and VW-PINNs in Fig. 15. The two results show good agreement, indicating our method accurately obtains the location of separation points and the scale of separation vortices.

For viscous incompressible flow, Rao et al. [45] suggested using the continuum-mechanics-based equations with a reduced order as the constraint in PINNs to enhance the accuracy. We employ this approach to further verify our perspective and evaluate the robustness of VW-PINNs. The continuum-mechanics-based equations are

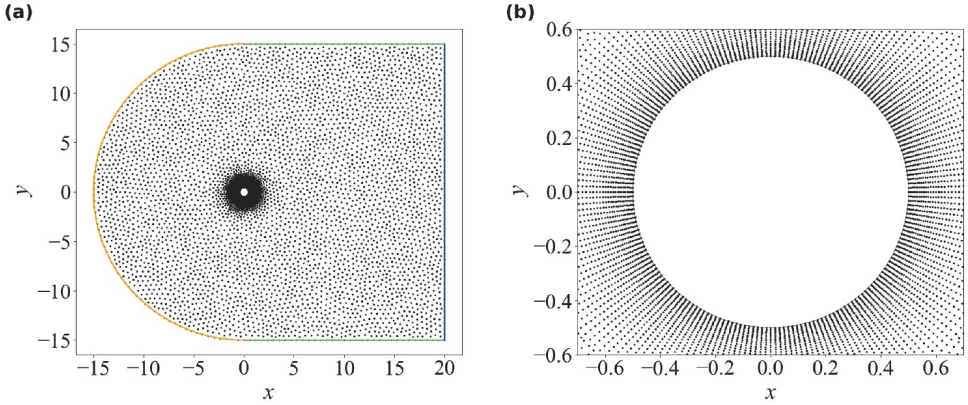


Figure 11 Computational domain and the distribution of collocation points in solving viscous incompressible flow over a circular cylinder. (a) Entire computational domain. The orange line represents the inflow boundary, the blue line represents the outflow boundary, and the green lines represent the upper and lower boundaries. (b) Near the cylinder region.

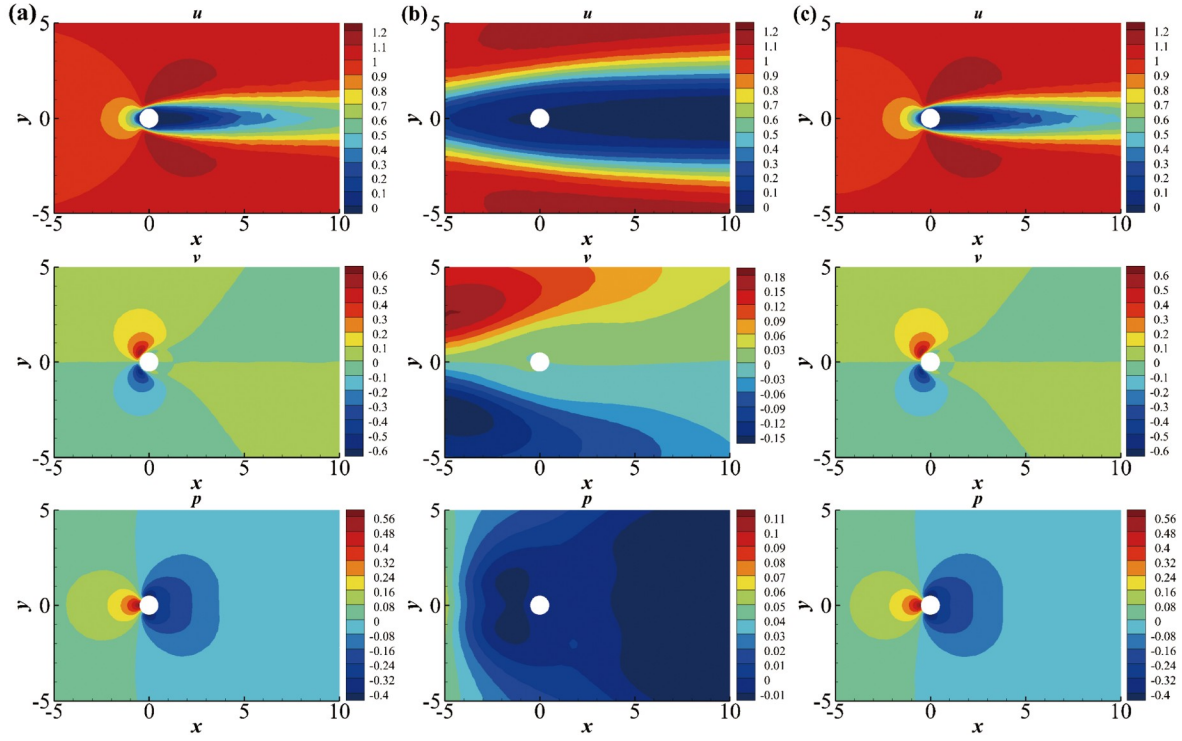


Figure 12 Comparison of the flow field near the cylinder among (a) the reference solution from the FVM, and the solutions given by (b) PINNs and (c) VW-PINNs for solving viscous incompressible flow over a circular cylinder.

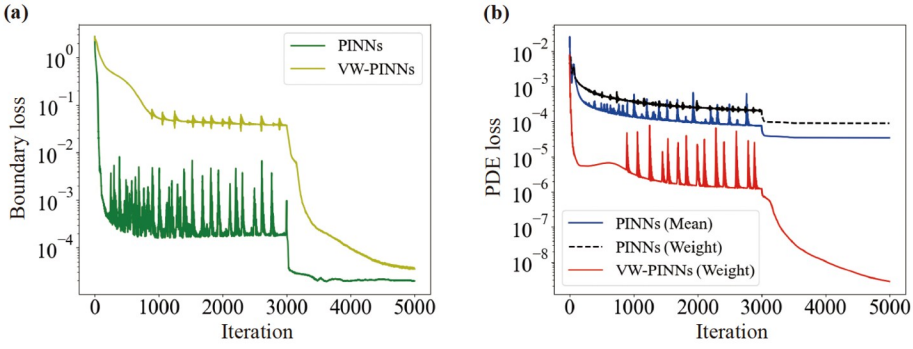


Figure 13 Comparison of loss function obtained by PINNs and VW-PINNs for solving viscous incompressible flow over a circular cylinder, respectively. (a) Boundary loss. (b) PDE loss.

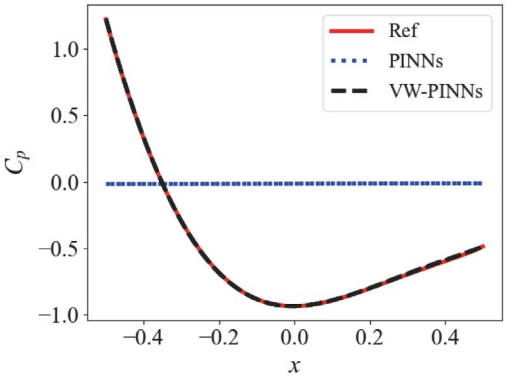


Figure 14 Comparison of pressure coefficient on the cylinder obtained by FVM (red), PINNs (blue) and VW-PINNs (black) for solving viscous incompressible flow over a circular cylinder, respectively.

$$\begin{aligned}
&\nabla \cdot \mathbf{V} = \mathbf{0},\\
&(\mathbf{V}\cdot\nabla)\mathbf{V} = \nabla\cdot\boldsymbol{\sigma},\\
&\boldsymbol{\sigma} = -p\mathbf{I} + \frac{1}{Re}(\nabla\mathbf{V} + \nabla\mathbf{V}^{\mathrm{T}}),\\
&p = -\mathrm{tr}\boldsymbol{\sigma}/2,
\end{aligned} \tag{16}$$

where ∇ is the Nabla operator, $\mathbf{V} = [u, v]$ is the velocity vector and $\boldsymbol{\sigma}$ is the Cauchy stress tensor (In two-dimensional problems, $\boldsymbol{\sigma}$ comprises three components). Consistent with Ref. [45], we define the stream function ψ, with $[x, y]$ as the network inputs and $[\psi, p, \boldsymbol{\sigma}]$ as the network outputs. $[u, v]$ can be calculated from ψ under incompressible condition:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \tag{17}$$

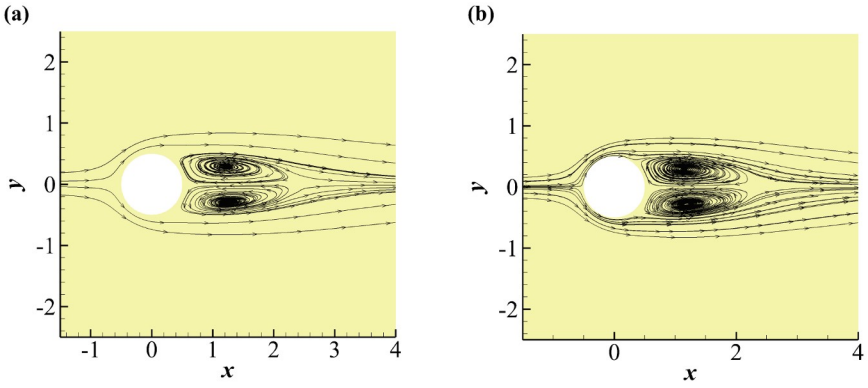


Figure 15 Comparison of streamline near the cylinder given by (a) FVM and (b) VW-PINNs for solving viscous incompressible flow over a circular cylinder, respectively.

Except that the number of neurons in the network output layer increases from 3 to 5 (due to $\boldsymbol{\sigma}$ comprising three components), the solving settings are the same as those used in solving standard Navier-Stokes equations (Eq. (15)). We solve the problem through PINNs and VW-PINNs respectively. Figure 16 shows the comparison of pressure coefficient on the cylinder obtained by the two methods. Similar to Fig. 14, the result obtained by VW-PINNs shows good agreement with the numerical solution, whereas the result obtained by PINNs is notably incorrect. This is reasonable, as according to our analysis in Sect. 2.2, the equation form is not the primary factor leading to the failure in solving problems involving nonuniform collocation points. The wall times of PINNs and VW-PINNs are 8.03 min and 20.18 min, respectively. The shorter wall time of PINNs is because they quickly converge to incorrect results and then encounter stagnation.

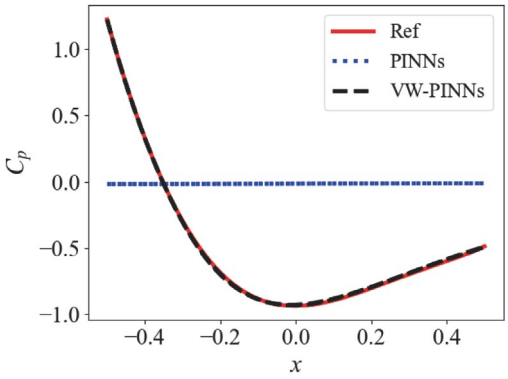


Figure 16 Comparison of pressure coefficient on the cylinder obtained by FVM (red), PINNs (blue) and VW-PINNs (black) for solving the continuum-mechanics-based equations, respectively.

3.3 Viscous incompressible flow over the NACA0012 airfoil

Our next example involves a scenario of the viscous incompressible flow over the NACA0012 airfoil. Compared with the cylinder, the curvature of the airfoil changes more rapidly. The governing equations of the problem are still Eq. (15). We consider the angle of attack $\alpha = 5°$, Reynolds number $Re = 400$, with $[x, y]$ as the network inputs and $[u, v, p]$ as the network outputs.

The leading edge point of the airfoil is located at $(0, 0)$ with chord $l = 1$. The computational domain and the distribution of collocation points are illustrated in Fig. 17. The total number of collocation points is $N_r = 14640$. The setting of boundary conditions is the same as Sect. 3.2. We solve the problem through PINNs and VW-PINNs respectively. The neural network contains 7 hidden layers with 64 neurons per layer, and tanh is chosen as the activation function. For the network training, we first run 3000 steps using the Adam optimizer, followed by an additional 10000 steps using the L-BFGS optimizer (maximum inner iterations is 20).

Figure 18 shows the comparison of the flow field near the airfoil among the reference solution, and the solutions given by PINNs and VW-PINNs. Similar to Sects. 3.1 and 3.2, VW-PINNs successfully solve viscous incompressible flow over the NACA0012 airfoil where PINNs fail. From the convergence of loss function, we observe that compared with PINNs, our approach reduces the volume weighting PDE loss by 3 additional orders of magnitude, as shown in Fig. 19. In Fig. 20, we evaluate the accuracy of the solving through the pressure coefficient on the airfoil. The relative L_2 error between the result obtained by VW-PINNs and the numerical solution is 3.21%, while PINNs obtain a wrong result. The wall times of PINNs and VW-PINNs are 22.36 min and 59.59 min, respectively. The shorter wall time of PINNs is because they quickly converge to incorrect results and then encounter stagnation.

3.4 Forward problem involving Burgers' equation based on adaptive sampling

In this section, we utilize the adaptive nonuniform sampling

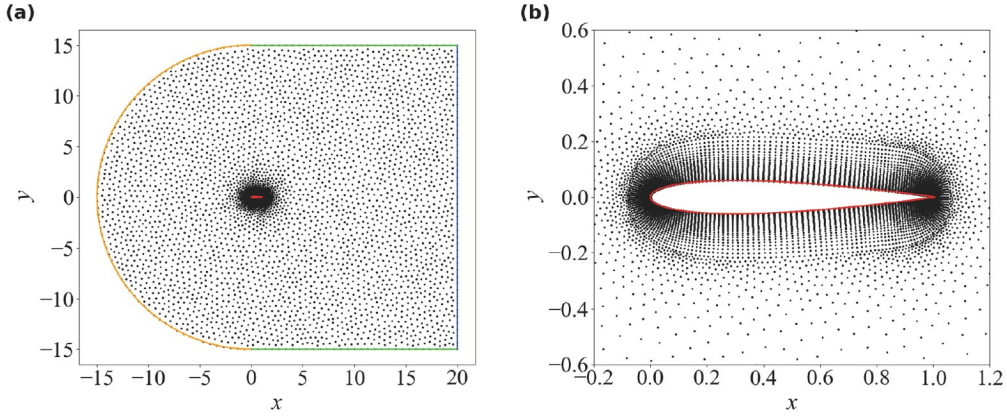


Figure 17 Computational domain and the distribution of collocation points in solving viscous incompressible flow over the NACA0012 airfoil. (a) Entire computational domain. The orange line represents the inflow boundary, the blue line represents the outflow boundary, the green lines represent the upper and lower boundaries, and the red line represents the airfoil. (b) Near the airfoil region.

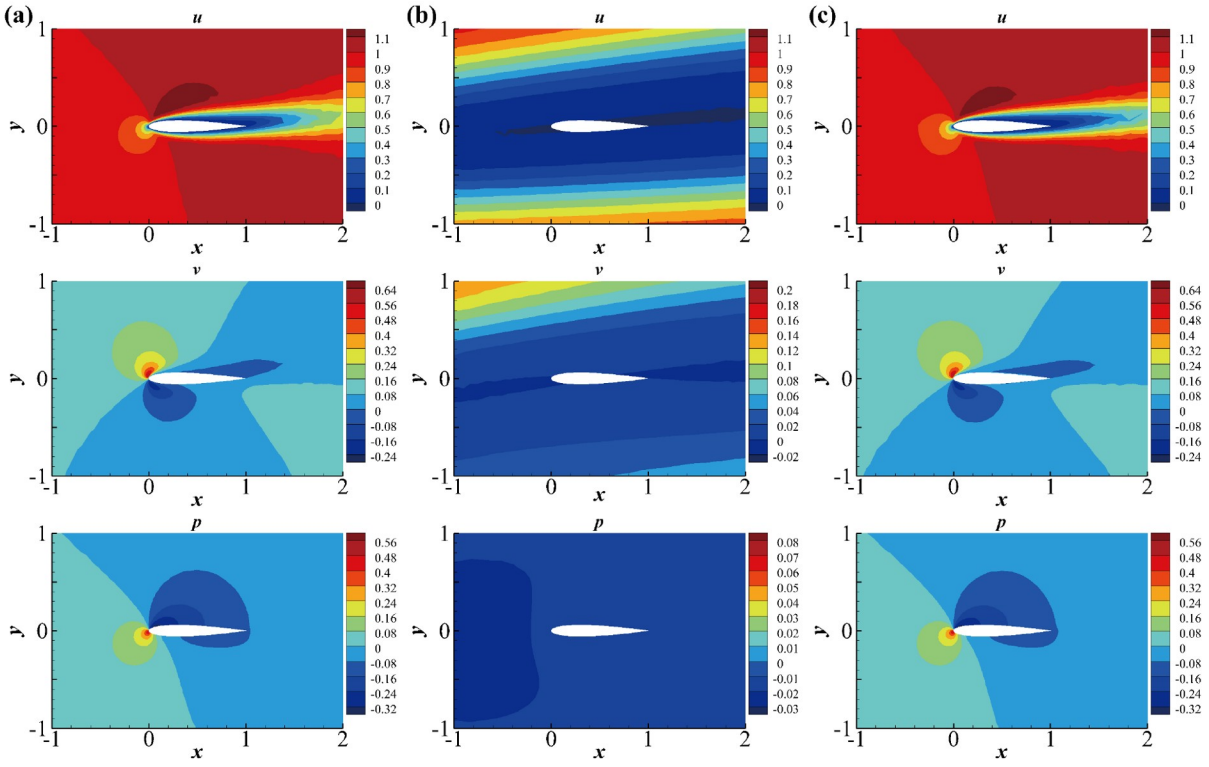


Figure 18 Comparison of the flow field near the airfoil among (a) the reference solution from the FVM, and the solutions given by (b) PINNs and (c) VW-PINNs for solving viscous incompressible flow over the NACA0012 airfoil.

method in PINNs proposed by Wu et al. [30] to validate the effectiveness of VW-PINNs in adaptive scenarios.

In Ref. [30], the authors proposed an adaptive sampling method for collocation points by evaluating PDE residuals during network training. Specifically, after predefined training steps I each time, the PDE residuals $\varepsilon(\mathbf{x}_s, t_s)$ are calculated on a highly dense set of test points $(\mathbf{x}_s, t_s)$ within the computational domain. Then, the probability density function $q(x_s, t_s)$ for adaptive sampling can be calculated by

$$q(\mathbf{x}_s, t_s) \propto \frac{\varepsilon^k(\mathbf{x}_s, t_s)}{\mathrm{E}[\varepsilon^k(\mathbf{x}_s, t_s)]} + c, \tag{18}$$

where $\mathrm{E}[\cdot]$ represents the mean operator. k and c are two hyperparameters, which determine the distribution of sampling points. According to the authors' descriptions, several other adaptive sampling methods [32-39] can all be summarized by Eq. (18), differing only in the values taken by k and c. Finally, the collocation points are obtained by sampling a subset of points from (x_s, t_s) according to $q(x_s, t_s)$.

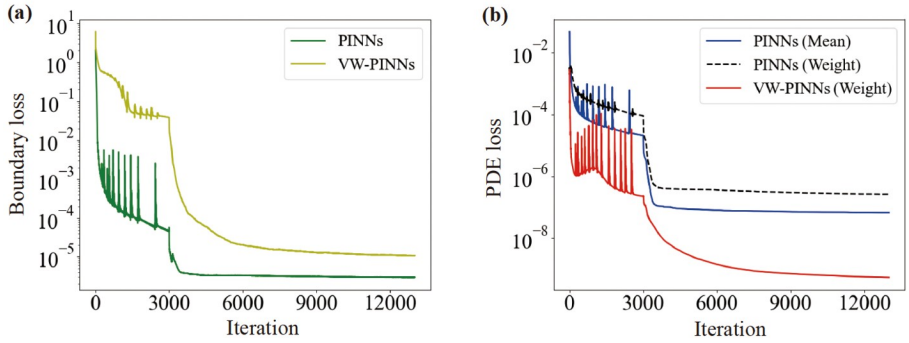


Figure 19 Comparison of loss function obtained by PINNs and VW-PINNs for solving viscous incompressible flow over the NACA0012 airfoil, respectively. (a) Boundary loss. (b) PDE loss.

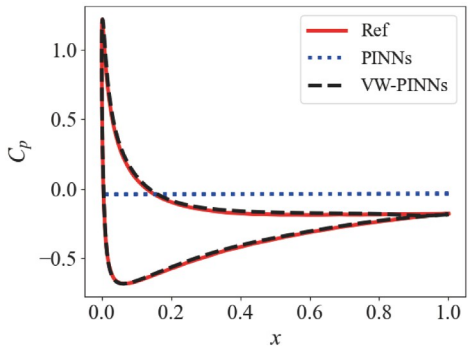


Figure 20 Comparison of pressure coefficient on the NACA0012 airfoil obtained by FVM (red), PINNs (blue) and VW-PINNs (black) for solving viscous incompressible flow over the NACA0012 airfoil, respectively.

Compared with uniform sampling, this method significantly enhances the solution accuracy of PINNs with sparse collocation points, especially when the solution of PDE has steep gradients.

We conduct the testing by solving Burgers' equation (Eq. (19)) in Ref. [30].

$$\begin{aligned} &\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}=\nu\frac{\partial^2 u}{\partial x^2},\quad x\in[-1,1],\quad t\in[0,1],\\ &u(x,0)=-\sin(\pi x),\\ &u(-1,t)=u(1,t)=0, \end{aligned} \tag{19}$$

where u is the flow velocity and $\nu=0.01/\pi$ is the viscosity of the fluid. The solution of the Burgers' equation has a sharp front when $x=0$ and t is close to 1, as shown in Fig. 21. We integrate algorithm based on the author's publicly available source code and use the same solving settings as the literature. The number of collocation points is fixed at $N_r=2000$, and definite conditions are imposed through hard constraints. The neural network contains 4 hidden layers with 64 neurons per layer, and tanh is chosen as the activation function. When solving the equation, firstly, based on uniform sampling, the neural network is trained using 15000 steps of Adam and then 1000 steps of L-BFGS. Subsequently, the adaptive sampling is employed (we use the best hyperparameter setting $k=1$, $c=1$ from the original literature). During each adaptive sampling period, the neural network is trained using 1000 steps of Adam and then 1000 steps of L-BFGS.

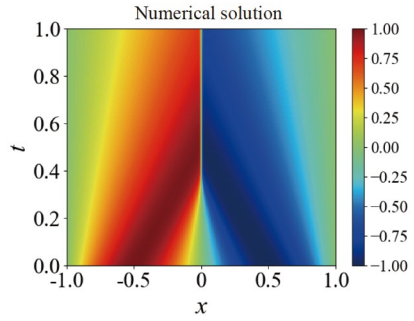


Figure 21 Numerical solution of Burgers' equation obtained by the spectral method.

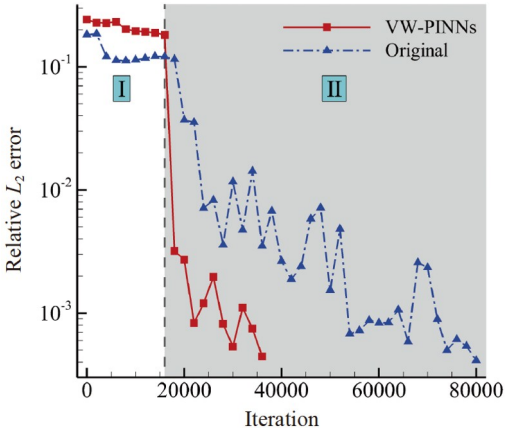


Figure 22 Relative L_2 error obtained by original method (blue) and VW-PINNs (red) for solving Burgers' equation, respectively.

Figure 22 shows the relative L_2 error obtained by the original method and VW-PINNs. The region I represents the training process based on uniform sampling, whereas region II corresponds to the adaptive sampling process. The result given by the original method is in good agreement with the result in Ref. [30]. When the volume weighting method is integrated, it is obvious that the solving of the new framework has a faster convergence speed. The wall times of the

original method and VW-PINNs are 7.27 min and 3.32 min, respectively, corresponding to a speedup of 2.19. If only the adaptive sampling training process is considered, the speedup reaches 3.20.

3.5 Inverse problem involving Burgers' equation

Solving inverse problems involving PDEs has been a major limitation of numerical methods [6]. PINNs can conveniently incorporate various observed data, enabling the solving of such problems at an extremely low cost. Therefore, we finally validate the VW-PINNs by solving the inverse problem involving Burgers' equation.

The governing equation of this problem is

$$\frac{\partial u}{\partial t}+\lambda_1 u\frac{\partial u}{\partial x}=\lambda_2\frac{\partial^2 u}{\partial x^2},\quad x\in[-1,1],\quad t\in[0,1], \tag{20}$$

where u is the flow velocity. In this problem, we aim to identify the unknown parameters λ_1, λ_2 and obtain a high-resolution u based on given observations from the numerical solution in Sect. 3.4. The exact values of the unknown parameter are $\lambda_1 = 1$ and $\lambda_2 = 0.01/\pi$.

The distributions of observed data and collocation points are shown in Fig. 23. We use 200 randomly observed data points. For the collocation points, according to u having a steep gradient at $x = 0$, we perform local refinement in this region, increasing the sampling density by five times than other regions, the total number is $N_r = 1000$. The network architecture follows the setting from Ref. [1], which contains 8 hidden layers with 20 neurons per layer, and tanh is chosen as the activation function. For the network training, we first run 2000 steps using the Adam optimizer, followed by an additional 6000 steps using the L-BFGS optimizer (maximum inner iterations is 1). The wall times of PINNs and VW-PINNs are 1.74 min and 1.76 min, respectively.

Figure 24 shows the relative errors of λ_1, λ_2 and u obtained by PINNs and VW-PINNs. We observe that the results of VW-PINNs are more accurate, especially in identifying the viscosity coefficient λ_2 with a relative error of only 1.48%, while PINNs yield an error of 19.82%. With improved equation identification, our method naturally achieves a more precise u. These results demonstrate the ability of VW-PINNs to solve inverse problems under nonuniform sampling.

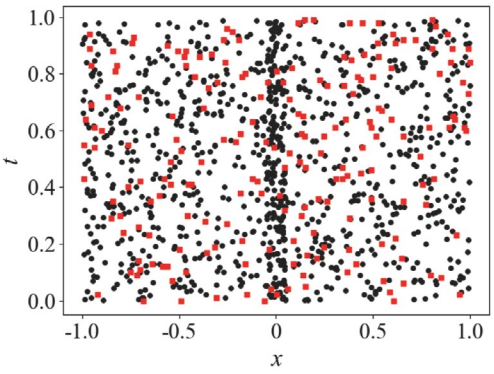


Figure 23 Distribution of sampling points in solving the inverse problem involving Burgers' equation. Black circles represent collocation points. Red squares represent observed data.

4. Conclusions

In this work, we first study the ill-conditioning of the PDE loss function in PINNs under nonuniform collocation points. Since the proportion of PDE residuals at all sampling points in the PDE loss is equal, when employing nonuniform sampling for collocation points, the loss reduction benefits brought by reducing the local residuals in high-density regions are higher than those in low-density regions. Consequently, the network will naturally focus on reducing residuals in high-density regions. In contrast, other low-density regions receive little attention and the residuals converge difficultly, resulting in poor PINNs performance. By solving the inviscid compressible flow over a circular cylinder, we analyze this problem specifically.

To address this issue, we define the volume weighting residual and propose VW-PINNs. Through weighting the PDE residuals by the volume that the collocation points occupy within the computational domain, we embed explicitly the distribution characteristics of collocation points in the PDE loss. Relying on the linear inverse relationship between the sampling density of collocation points and the volume they occupy, VW-PINNs balance the differences in

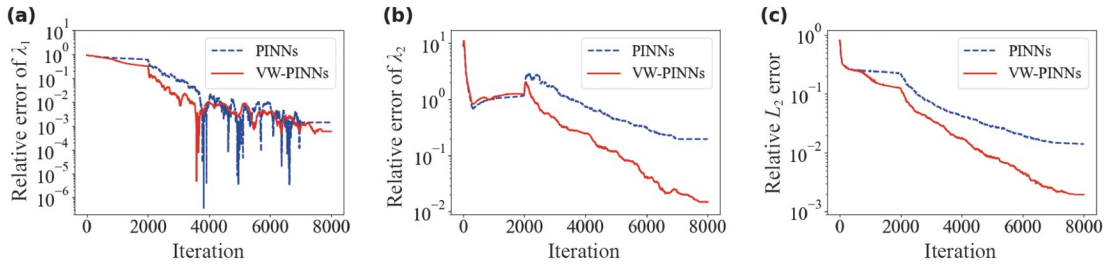


Figure 24 Relative error in solving the inverse problem involving Burgers' equation. (a) λ_1. (b) λ_2. (c) Evaluating u at a set of dense test points (the number is 2.56×10^4).

residual convergence across regions with various sampling densities in network optimization. The reduction of the volume weighting PDE loss ensures efficient convergence of the total PDE residual across the entire computational domain. To calculate the volume of collocation points, we also develop a volume approximation algorithm suitable for random sampling based on the kernel density estimation method. It only requires the coordinates of sampling points. By solving four forward problems and one inverse problem, we validate the effectiveness of the proposed method:

(1) VW-PINNs successfully solve the flow over a circular cylinder and the flow over the NACA0012 airfoil under different inflow conditions, while conventional PINNs fail. The relative L_2 errors of the wall pressure coefficient are only about 1%, and the flow separation behind the cylinder is accurately captured.

(2) By integrating the volume weighting method into the existing adaptive sampling approach, we triple the solving efficiency of the adaptive sampling approach.

(3) In solving the inverse PDE problem with nonuniform sampling, compared with PINNs, VW-PINNs reduce the relative error by one order of magnitude.

However, our method still faces challenges in solving flow at high Reynolds numbers and flow over the airfoil under compressible conditions. These challenges may be solved in the future by using recent advances.

Conflict of interest *On behalf of all authors, the corresponding author states that there is no conflict of interest.*

Author contributions ***Jiahao Song:*** *Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft.* ***Wenbo Cao:*** *Formal analysis, Investigation, Software, Writing – review & editing.* ***Fei Liao:*** *Methodology, Supervision.* ***Weiwei Zhang:*** *Conceptualization, Funding acquisition, Supervision, Writing – review & editing.*

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VW-PINNs: 物理信息神经网络中PDE残差的体积加权方法

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摘要 物理信息神经网络(Physics-informed neural networks, PINNs)已经在偏微分方程(Partial differential equations, PDEs)正反问题求解领域展示出广阔前景. 该方法通过在一组配置点上计算PDE损失实现了偏微分方程在神经网络中的嵌入, 具有无网格、易于自适应采样等优势. 然而, 当使用非均匀配置点求解方程时, PINNs面临方程残差收敛效率低甚至求解失败的问题. 本研究首先分析了非均匀配置点下PINNs中PDE损失函数的病态. 然后针对性地定义了体积加权残差的概念, 并提出了体积加权的物理信息神经网络(Volume weighting physics-informed neural networks, VW-PINNs). 通过使用配置点在求解区域占据的体积对方程残差进行加权, 配置点的空间分布特征被显式地引入PDE损失函数中, 保证了方程残差快速充分的收敛. 同时, 考虑到VW-PINNs的无网格特性, 我们还基于核函数估计发展了近似算法来计算配置点的体积. 我们通过求解PINNs失效的不同来流条件下的圆柱、翼型绕流正问题验证了VW-PINNs的普适性. 通过求解Burgers'方程, 我们验证了VW-PINNs能够将现有自适应采样方法在正问题上的收敛速度提升三倍, 能够将PINNs在反问题上的相对L_2误差降低一个数量级以上.